

Activity-Based Costing Practice Problem

Precision Outdoor Equipment

Precision Outdoor Equipment manufactures two products: **Standard Trimmers** and **Professional Trimmers**. The company currently applies manufacturing overhead using a single plantwide predetermined overhead rate based on direct labor-hours.

Current Plantwide Costing System

For the upcoming year, the company expects the following manufacturing overhead costs:

Manufacturing Overhead	Annual Cost
Machine-related costs	\$360,000
Production setup costs	\$180,000
Material handling costs	\$120,000
Total manufacturing overhead	\$660,000

The company expects to use **30,000 direct labor-hours** during the year.

The following production and direct-cost information is available. Direct materials and direct labor are traced directly to each product and do not change when the overhead allocation method changes.

	Standard Trimmer	Professional Trimmer
Units produced	10,000	2,000
Direct materials per unit	\$48	\$85
Direct labor per unit	\$30	\$60
Direct labor-hours per unit	1.5	3.0

Part A - Plantwide Overhead Rate

1. Calculate the predetermined manufacturing overhead rate.
2. Calculate the manufacturing overhead assigned to each product.
3. Calculate manufacturing overhead per unit for each product.
4. Calculate the **total manufacturing cost per unit** for each product: direct materials + direct labor + applied manufacturing overhead.

Proposed Activity-Based Costing System

Management believes the plantwide allocation system may be distorting product costs. It is considering an activity-based costing system with the following activity cost pools:

Activity Cost Pool	Estimated OH	Cost Driver	Estimated Activity
Machine processing	\$360,000	Machine-hours	18,000 MH
Production setups	\$180,000	Number of setups	90 setups
Material handling	\$120,000	Material moves	600 moves
Total	\$660,000		

Expected activity by product:

	Standard Trimmer	Professional Trimmer
Machine-hours	12,000	6,000
Number of setups	30	60
Material moves	200	400

Part B - Activity-Based Costing

1. Calculate an activity rate for each cost pool.
2. Assign manufacturing overhead to each product using activity-based costing.
3. Calculate manufacturing overhead per unit for each product.
4. Calculate the **total manufacturing cost per unit** under ABC: direct materials + direct labor + assigned manufacturing overhead.
5. Compare the total cost per unit under ABC with the total cost per unit using the plantwide overhead rate.